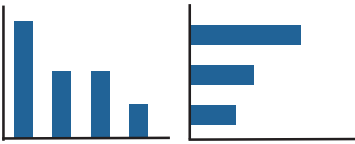
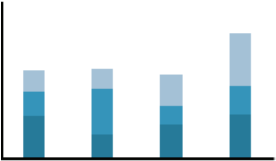
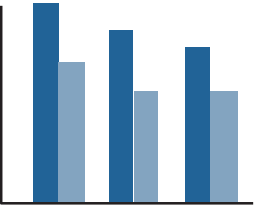
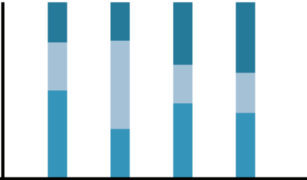
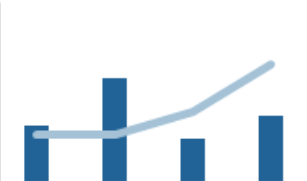


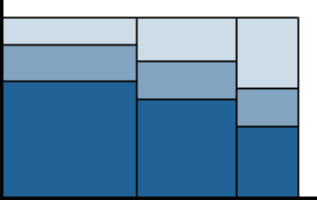
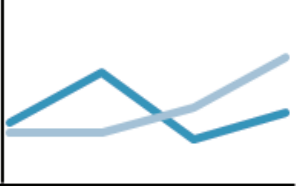
Communicating Quantitative Insights

CHART TYPE	OVERVIEW	TIPS AND CONSIDERATIONS
Bar or Column 	• Compare quantities across different categories • Uses rectangular bars with lengths proportional to the values they represent	• Versatile • Easy to visually process • Limited to discrete data • Can become cluttered with too many categories • Order of segments matters in relaying take aways • Time series moves left to right • Spot color is valuable for highlighting key points
Stacked Bar 	• Shows parts of a whole across different categories or over time • Bars are divided into segments representing different subcategories, stacked on top of each other	• One of the most used and powerful chart types • Best used for segmented data compared over time • Can see absolute and relative size of segment over time • Hard to interpret with many segments • Hard to perceive changes in small segments
Side-by-Side Bar 	• A bar chart where bars for different categories are placed next to each other for comparison • Often referred to as a clustered bar	• Displays absolute numbers • Allows side-by-side comparison of multiples variables • When comparing data over time, works best for two variables. Time series stacked bars may be better alternative for more variables. • Can become cluttered with too many categories • Best for comparative data that is discrete

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CHART TYPE	OVERVIEW	TIPS AND CONSIDERATIONS
100% Bar 	- Shows relative proportions (parts of a whole) within categories, normalized to 100% - Each bar represents 100% and is divided into segments representing subcategories	- Better option than a pie chart for showing share within a population - Multiple 100% bars can show how that population changes over time - Does not show absolute size of the population in each year, which might imply that segments are proportional; bar totals can help avoid this - Segment lines help track changes and can show increases or decreases
Combo 	- Compares different datasets with varying scales on the same chart - Uses two y-axes to display datasets with different scales	- Allows comparison of and the display of the relationship between two different types of variables - Maximizes data density - Sales typically shown as a bar (on left axis), and some sort of driver (e.g., marketing budget) shown as the line (on right axis) - May cause confusion with multiple scales
Waterfall 	- Provides a visual map from one value to another - Shows cumulative effects of sequential positive and negative adjustments	- Highlights the most powerful drivers of change - Data can be categorical, such as revenue and expense types, or can show movement of a variable over time - Useful for visualizing financial data, share prices, profit and loss statements, or sales volumes. - Can be a difficult analysis to do and needs solid data - Can be difficult to create without special software (e.g., think-cell)

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CHART TYPE	OVERVIEW	TIPS AND CONSIDERATIONS
Mekko 	• Variable width bar charts, showing data with varying magnitudes • Represents two variables, where the width of each bar indicates one variable, and the height indicates another	• Both X and Y axis convey relative size of segments, thus the area of any given segment can be compared to any other area • Often used to show measures like growth, margin or market share by product, segment or region • Useful showing market growth by segment where growth rate is the bar height and segment revenue is the bar width • May need detailed explanation for some audiences
Line 	• Shows data points connected by straight lines • Consists of x-axis (typically time) and y-axis (variable being measured)	• Useful for showing trends over time • Commonly used for key performance metrics, such as revenue growth, market trends, customer acquisition rates, and operational efficiency. • Helpful in visualizing trends and forecasting future performance
Area 	• A line chart with the area below the line filled in, representing cumulative totals	• Highlights individual trends and overall pattern • Useful for time series data • Can be difficult to interpret with overlapping data • Limited to continuous data

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CHART TYPE	OVERVIEW	TIPS AND CONSIDERATIONS
Scatter Plot 	- Shows relationships between two variables - Uses dots to represent values for two different variables, with the position of each dot determined by the values.	- Aids in pattern recognition and outliers - Useful for correlation analysis - Potential for data point clutter - Risk of subjective interpretation
Bubble Chart 	A scatter plot where the size of each bubble represents a third variable, and the color represents additional variables	- Useful for complex data and highlight relationships between three or more variables - Not effective when additional dimensions don't add enough incremental value to make up for their presence
Tornado Chart 	- Visualized the impact of various risks or uncertainties on a project's outcome - Y-axis lists the variable you want to measure and the x-axis shows impact	- Clear visualization of which variables have most significant impact - Useful for evaluating market changes, competition, supply chain reliability, and other drivers - Single variable focus - Limited context; doesn't capture interaction between multiple factors